obstacle to it hosting a tiny website. So instead of using a sledgehammer to crack a nut, what you need to do is look for an alternative.

Among the several lightweight Web servers available is Lighttpd, pronounced 'lighty'. Its homepage is at *http://www.lighttpd.net/*. Lighttpd has all the features required of a modern Web server, including FastCGI, authentication and URL rewriting, but makes do with modest resources, and is suitable for use on relatively resource-constrained systems. This article demonstrates how to install and configure Lighttpd on a Fedora system; the procedure for other distributions is similar.

Using your package-management system, install the *lighttpd* package. On Fedora systems, you also need to install the *lighttpd-fastcgi* package to be able to run PHP. Later, we will look at how to set up the server to run PHP.

Configuring Lighttpd

Lighttpd uses a single primary configuration file named *lighttpd.conf*, which is stored in */etc/lighttpd/*. Open this file in a text editor with root privileges. The configuration parameters in this file are well-commented, and also have self-explanatory names. The *server.modules* parameter is an array of modules to be loaded by Lighttpd when it starts. At the very least, *mod_access* and *mod_accesslog* should be enabled. For our purposes, we also need to enable *mod_fastcgi*. To get a more comprehensive feel for Lighttpd configuration, also enable *mod_userdir*, which will enable you to use a subdirectory of your home directory to serve Web pages. In the *server.document-root* parameter, provide the path to a default document root for your server (not the subdirectory that is to be used through *mod_userdir*). In the *index-file.names* array, include *index.php* or *default.php* or whatever file name(s) you use for your default PHP pages, if not already present. Lighttpd allows you to selectively prevent or allow access to specific files or directories. For example, to prevent access to directories with *_private* in the name, use the following configuration parameter:

```
$HTTP["url"] =~ "_private" {
  url.access-deny = ("")
}
```

Next, make sure that the *static-file.exclude-extensions* array contains the '*.php*', '*.pl*' and '*.fcgi*' extensions, to enable running PHP, Perl and other FastCGI programs. If you use other extensions for your programs, add them here. If you want to run Lighttpd along with another Web server, specify the TCP port that Lighttpd should listen on for incoming requests, in the server.port parameter (it is commented out by default).

The major part of the configuration to run PHP is the fastcgi.server parameter. This should look like what follows:

```
fastcgi.server = ( ".php" =>
                   ( "localhost" =>
                     (
                       "socket" => "/var/run/lighttpd/php-
fastcgi.socket",
                       "bin-path" => "/usr/bin/php-cgi"
                     )
                   )
                 )
```

Finally, ensure that there is an include_shell command near the end of the file including files within */etc/lighttpd/conf.d/*. Don't try to run PHP yet.

Since we had enabled *mod_userdir*, it is time to provide a *userdir* configuration. In the */etc/lighttpd/conf.d/* directory, create a text file named *userdir.conf* with root ownership, if it does not exist. In that file, provide just a single parameter, *userdir.path*, which specifies the subdirectory of your home directory from which you want to serve files. The path is relative to your home directory—for example, *userdir.path = "public_html"* or *userdir.path = "Projects/www"*.

The configuration of Lighttpd is now complete. The next step is to set up PHP to work with it.

Setting up PHP to work with Lighttpd

PHP is usually called as a module in the Apache Web server. However, with Lighttpd, it needs to be called as a FastCGI program. To enable this, PHP has a version of the interpreter that runs as a standalone program. On Fedora, install the *php-cli* package for it; there may be other packages for other systems. (Note: the package named *php* on Fedora provides the Apache module for PHP, which is not what you want.)

After the PHP standalone program is installed, the *cgi.fix_pathinfo* PHP configuration parameter needs to be set for PHP to work correctly with Lighttpd. On Fedora, you should have a file named *lighttpd.ini* in */etc/php.d/* after the *php-cli* installation. If not, create it with *root* ownership. It should just have: *cgi.fix_pathinfo = 1*. Alternatively, or on other systems, if you do not want or have *lighttpd.ini* in */etc/php.d/*, you can set this parameter within the main *php.ini* file itself.

At this point, you can run Lighttpd and also run PHP programs through it. To start Lighttpd on Fedora, use the command *su -c 'service lighttpd start'* and enter your root password when prompted. To stop Lighttpd, just use the word *stop* instead of *start* in that command; use *status* to get the service's current status. The *sudo* command can also be used for the purpose, if you have *sudo* privileges.

Accessing files using *mod_userdir* is similar to the way it is done in Apache—construct the file's URL using *~username*. For example, if I specify '*Projects/www*' in the *userdir.path* parameter and want to run *test.php* stored